final_project_code

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1. Reading in the data

```
read_data <- read_excel("./stat571B_reading_data.xlsx")

read_data['total_time'] = read_data['Time (minutes)']*60 + read_data['Time (seconds)']

read_data['format'] = c("Laptop", "Laptop", "Print",
                        "Laptop", "Print", "Print",
                        "Print", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Laptop", "Print", "Print",
                        "Laptop", "Print", "Laptop",
                        "Print", "Laptop", "Laptop",
                        "Print", "Laptop", "Print",
                        "Laptop", "Print", "Laptop",
                        "Laptop", "Laptop", "Print",
                        "Print", "Laptop", "Print",
                        "Print", "Print", "Laptop")

read_data["music"] = c("no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "yes", "no",
                       "yes", "no", "yes",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "no", "yes",
                       "yes", "yes", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no",
                       "no", "yes", "yes",
                       "yes", "no", "no")

read_data
```

```
## # A tibble: 36 x 8
##    Participant  Text `Comprehension score (%)` `Time (minutes)` `Time (seconds)`
##    <chr>       <dbl>                     <dbl>            <dbl>            <dbl>
##  1 P1              1                       100                2               17
##  2 P1              2                        75                2               45
##  3 P1              3                        75                1               58
##  4 P2              1                        75                3               14
##  5 P2              2                        50                2               36
```

```
##  6 P2              3                        75                2                51
##  7 P3              1                        75                2                 5
##  8 P3              2                       100                2                28
##  9 P3              3                        75                1                49
## 10 P4              1                        75                2                22
## # i 26 more rows
## # i 3 more variables: total_time <dbl>, format <chr>, music <chr>
```

2. Analysis

Model 1: Time

First we fitted a full factorial linear model including format, music, text, and participant as fixed effect to explore the overall patterns in the data. An ANOVA was created and provided a significant main effect of music (p=0.04867), text (p=0.02744), and participants (p=0.01437). This is something we expected before analyzing. Also, we saw that no interaction effects were significant.

However, because participants and text represent a random sample, we then fitted a mixed-effects model treating participants and text as our random effects. We also wanted to explore participant specific random slope for music. Changing our model to this allows for results to be generalized beyond our specific data collection.

From the ANOVA of this model, we concluded that music was the only statistically significant main effect with a p-value of 0.0112. Displayed below is a bar plot showing the effect of music on reading time. We see that for all text reading time increased with music usage.

```r
# ANOVA for Reading Time
model_time <- lm(total_time ~ format * music + Text + Participant + Participant:music,
                 data = read_data)
Anova(model_time, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: total_time
##                    Sum Sq Df  F value    Pr(>F)
## (Intercept)       31068.0  1 100.1084 8.454e-06 ***
## format              374.1  1   1.2054   0.30419
## music              1675.1  1   5.3975   0.04867 *
## Text               3617.3  2   5.8279   0.02744 *
## Participant       17428.3 11   5.1053   0.01437 *
## format:music          0.0  1   0.0001   0.99104
## music:Participant  7061.3 11   2.0685   0.15580
## Residuals          2482.7  8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Mixed Effects Model
model_time_lmer <- lmer(
  total_time ~ format * music +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

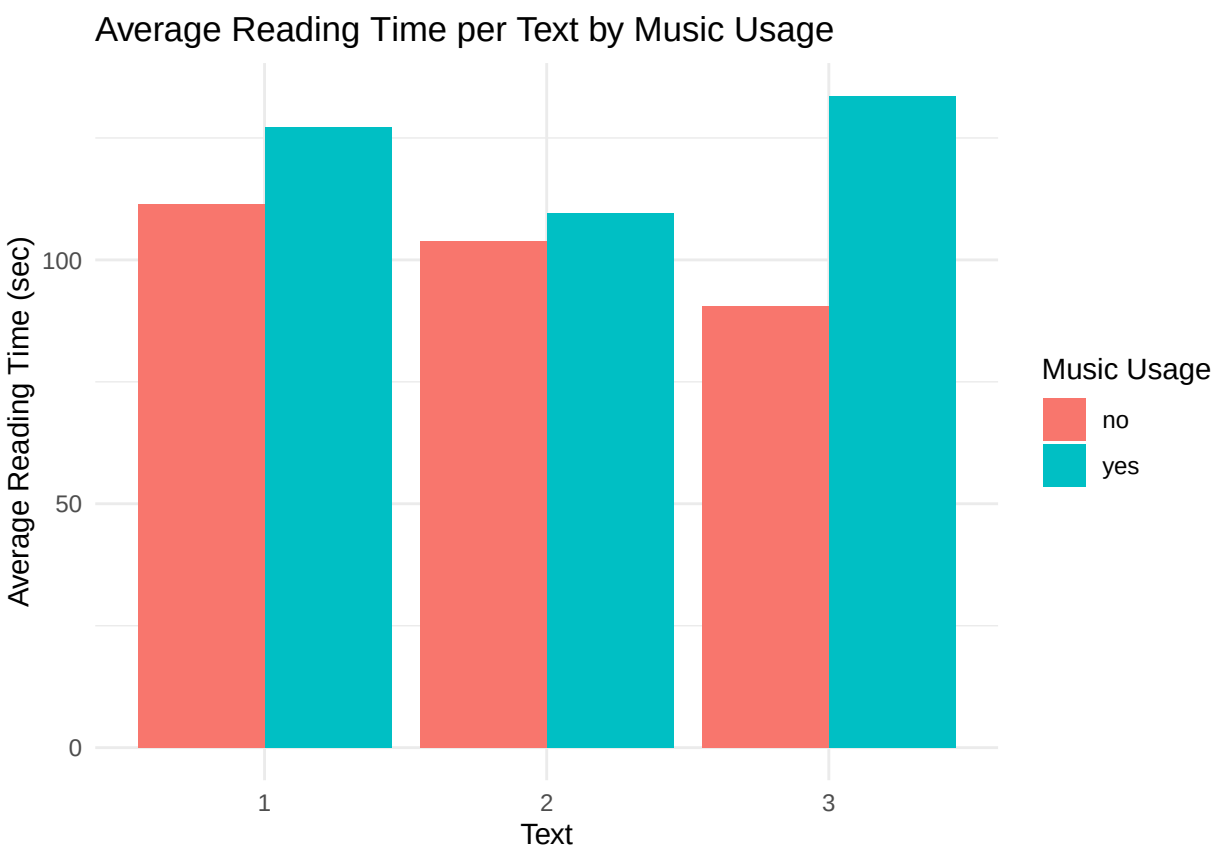
```r
anova(model_time_lmer)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##              Sum Sq Mean Sq NumDF  DenDF F value Pr(>F)
## format        340.4   340.4     1 21.932  0.7002 0.4117
## music        3854.8  3854.8     1 18.570  7.9289 0.0112 *
## format:music   12.0    12.0     1 21.932  0.0246 0.8768
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Average time per Text split by Music

avg_time_text_music <- read_data %>%
  group_by(Text, music) %>%
  summarise(avg_time = mean(total_time, na.rm = TRUE), .groups = "drop")

# Bar plot comparison
ggplot(avg_time_text_music, aes(x = Text, y = avg_time, fill = factor(music))) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Average Reading Time per Text by Music Usage",
    x = "Text",
    y = "Average Reading Time (sec)",
    fill = "Music Usage"
  ) +
  theme_minimal()
```



Model 2: Comprehension

First we fitted a full factorial linear model including format, music, text, and participant as fixed effect to explore the overall patterns in the data. An ANOVA was created and provided no significant effect on comprehension for any main effect. We believe this is due to the lack of variance in our comprehensions scores.

Again, because participants and text represent a random sample, we then fitted a mixed-effects model treating participants and text as our random effects. From this we concluded only the interaction between format and music showed statistical significant effect on comprehension with a p-value of 0.03796. No main effect significance is also evidence for us to believe there is some lack of variance in our comprehensions scores.

```r
# ANOVA for Comprehension
model_comp <- lm(`Comprehension score (%)` ~ format * music + Text +
                   Participant + Participant:music,
                  data = read_data)
Anova(model_comp, type = "III")
```

```
## Anova Table (Type III tests)
##
## Response: Comprehension score (%)
##                     Sum Sq Df F value    Pr(>F)
## (Intercept)       12604.2  1 52.3243 8.947e-05 ***
## format              833.3  1  3.4595   0.09994 .
## music               400.1  1  1.6609   0.23349
## Text                364.6  2  0.7568   0.50003
## Participant        3755.8 11  1.4174   0.31702
## format:music        416.7  1  1.7297   0.22489
## music:Participant  3281.3 11  1.2383   0.38986
## Residuals          1927.1  8
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
## Mixed Effects Model
model_comp_lmer <- lmer(
  `Comprehension score (%)` ~ format * music +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
anova(model_comp_lmer)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF  DenDF F value  Pr(>F)
## format         65.72   65.72     1 24.289  0.2764 0.60384
## music           1.28    1.28     1 20.002  0.0054 0.94230
## format:music 1145.62 1145.62     1 24.289  4.8179 0.03796 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Since the comprehension model does not lead to very interesting analysis of results, we have decided to fit another model treating it as a fixed effect in order to determine if it had any effect on reading time. We would expect that lower comprehension score would be correlated with lower reading time. We must acknowledge however, that this is not a causal relationship but can be indicative of a relationship between these variables.

From this model again we concluded no main effect was statistically significant. Although comprehension is not a statistically significant predictor, the scatter plot with a regression line indicates a negative association between comprehension score and reading time.

```r
model_time_lmer <- lmer(
  total_time ~ format * music * `Comprehension score (%)` +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
anova(model_time_lmer)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                          Sum Sq Mean Sq NumDF  DenDF F value
## format                                   129.85  129.85     1 19.742  0.2417
## music                                    603.26  603.26     1 19.337  1.1227
## `Comprehension score (%)`                914.05  914.05     1 20.358  1.7011
## format:music                            1185.60 1185.60     1 20.600  2.2065
## format:`Comprehension score (%)`         197.38  197.38     1 19.893  0.3673
## music:`Comprehension score (%)`          203.39  203.39     1 19.070  0.3785
## format:music:`Comprehension score (%)` 1058.63 1058.63     1 20.903  1.9702
##                                         Pr(>F)
## format                                  0.6284
## music                                   0.3024
## `Comprehension score (%)`               0.2067
## format:music                            0.1526
## format:`Comprehension score (%)`        0.5513
## music:`Comprehension score (%)`         0.5457
## format:music:`Comprehension score (%)` 0.1751
```

```r
# last model
model_score_lmer <- lmer(
  total_time ~ format * music * `Comprehension score (%)` +
    (1 | Text) +
    (1 + music | Participant),
  data = read_data
)
```

```
## boundary (singular) fit: see help('isSingular')
```

```r
anova(model_score_lmer)
```

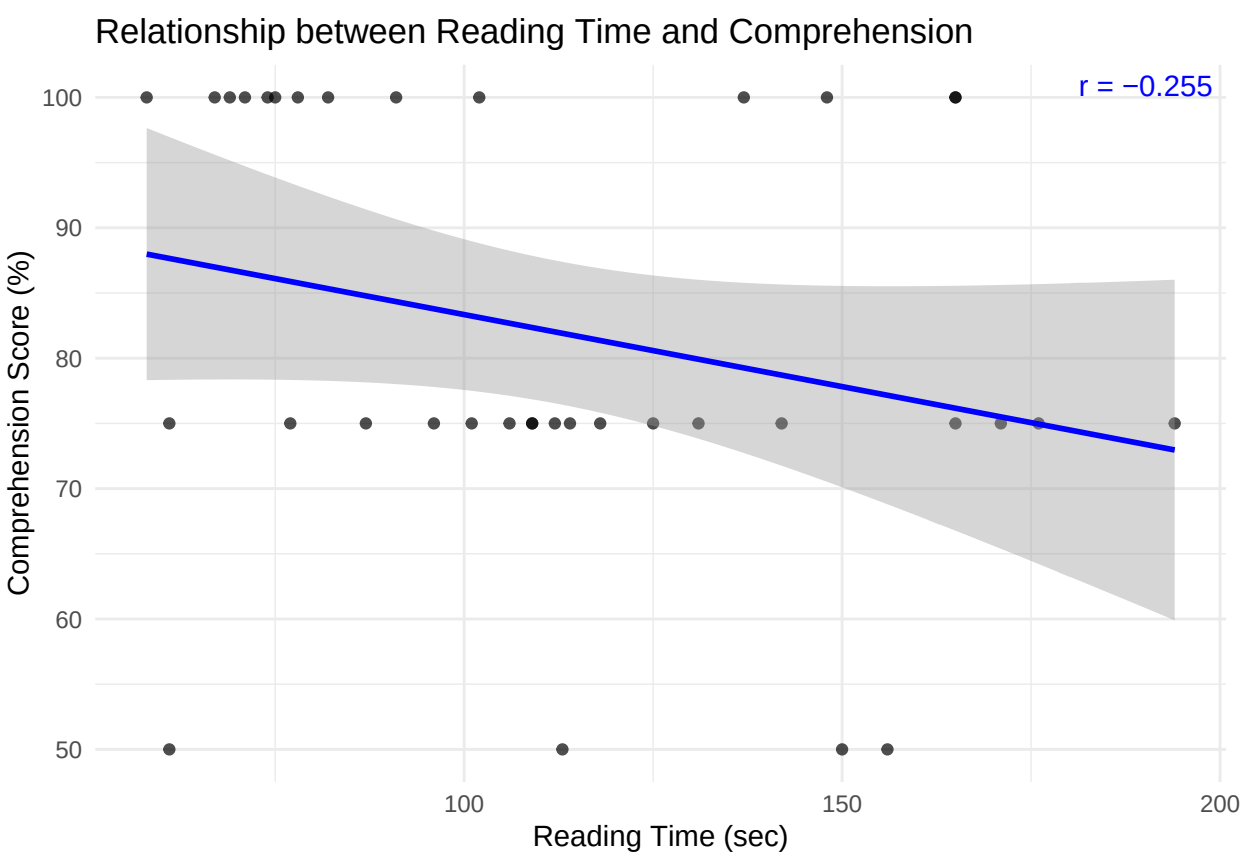
```
## Type III Analysis of Variance Table with Satterthwaite's method
##                                          Sum Sq Mean Sq NumDF  DenDF F value
## format                                   129.85  129.85     1 19.742  0.2417
## music                                    603.26  603.26     1 19.337  1.1227
## `Comprehension score (%)`                914.05  914.05     1 20.358  1.7011
## format:music                            1185.60 1185.60     1 20.600  2.2065
## format:`Comprehension score (%)`         197.38  197.38     1 19.893  0.3673
## music:`Comprehension score (%)`          203.39  203.39     1 19.070  0.3785
## format:music:`Comprehension score (%)` 1058.63 1058.63     1 20.903  1.9702
##                                         Pr(>F)
## format                                  0.6284
## music                                   0.3024
```

```
## `Comprehension score (%)`              0.2067
## format:music                           0.1526
## format:`Comprehension score (%)`       0.5513
## music:`Comprehension score (%)`        0.5457
## format:music:`Comprehension score (%)` 0.1751
```

```r
#correlation
cor_val <- cor(read_data$total_time, read_data$`Comprehension score (%)`)

# relationship plot
ggplot(read_data, aes(x = total_time, y = `Comprehension score (%)`)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, color = "blue") +
  annotate("text",
           x = Inf, y = Inf,
           label = paste("r =", round(cor_val, 3)),
           hjust = 1.1, vjust = 1.5,
           color = "blue") +
  labs(
    title = "Relationship between Reading Time and Comprehension",
    x = "Reading Time (sec)",
    y = "Comprehension Score (%)"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



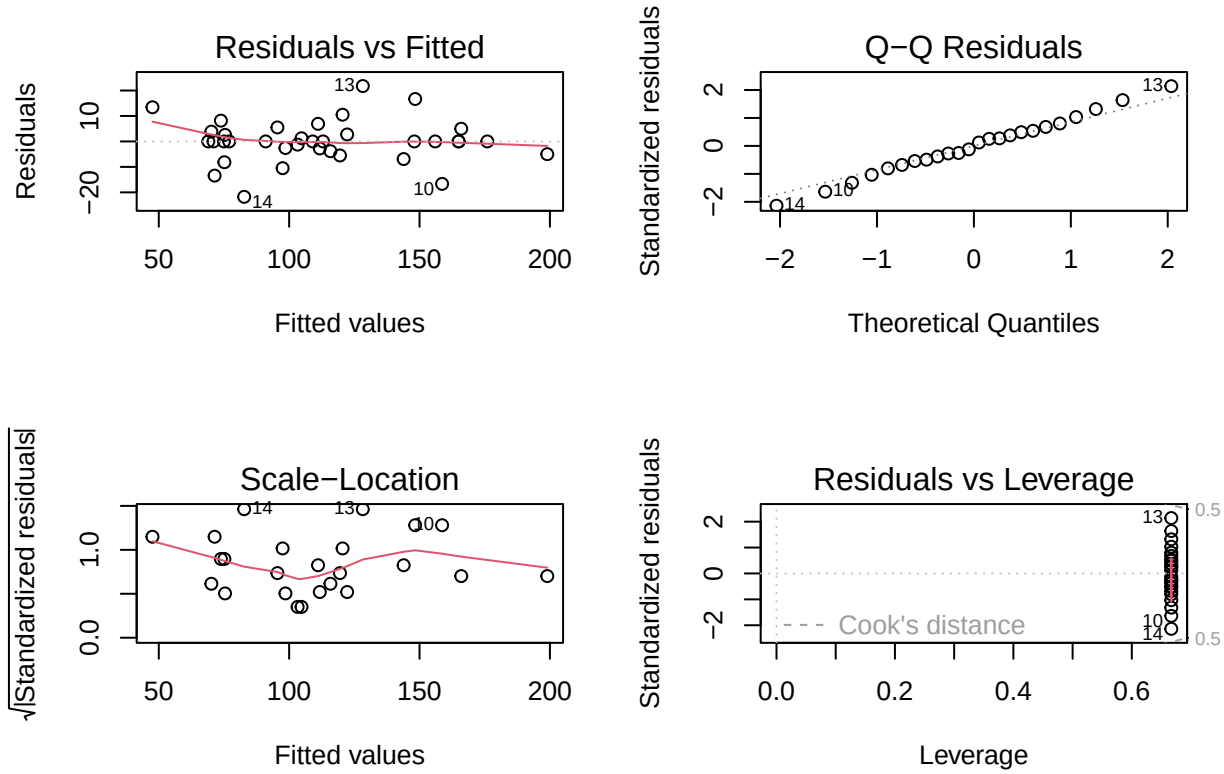
From this analysis, we can conclude that there was insufficient evidence to detect a significant effect on

reading comprehension. However, we did find that music has an significant effect on total reading time. With more time for data collection and a larger sample size, it would be interesting to investigate whether additional factors influence these responses or whether stronger effects emerge. Overall, this model gives a great starting point for further exploration of these relationships and could lead to more interesting findings.

Diagnostics for ANOVA

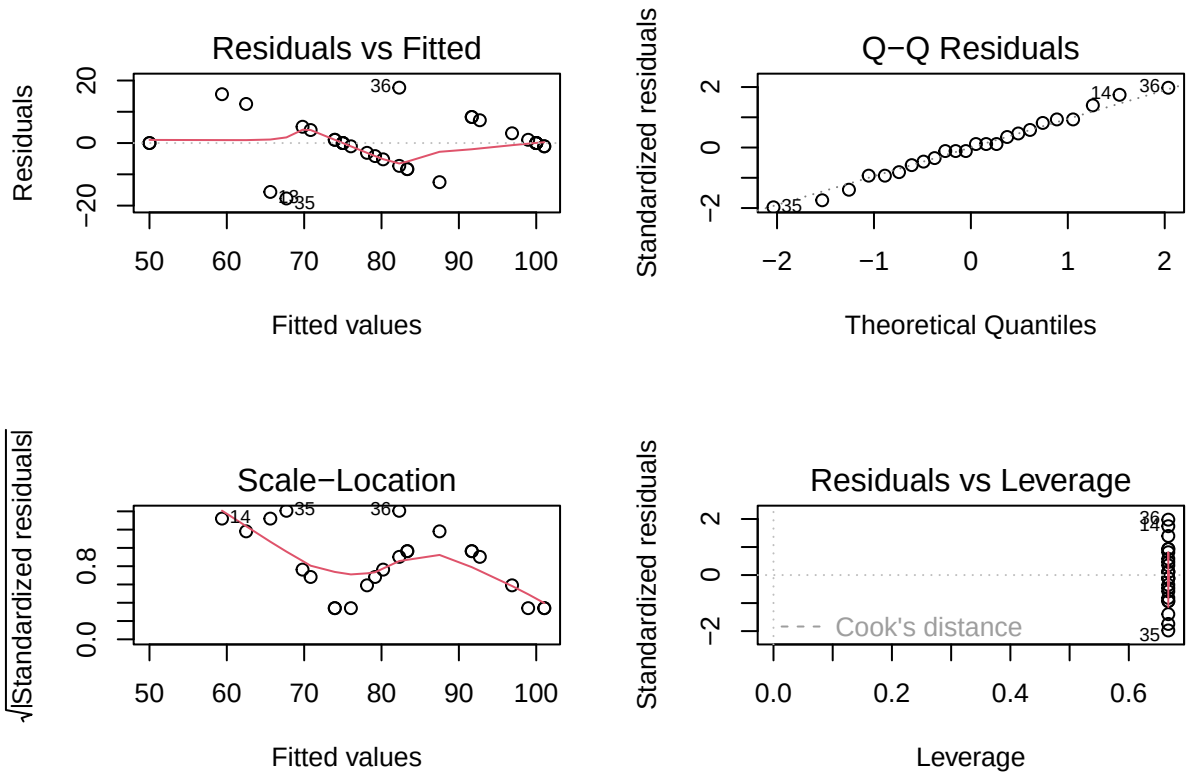
```
par(mfrow = c(2,2))
plot(model_time)
```

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```




```
par(mfrow = c(2,2))
plot(model_comp)
```

```
## Warning: not plotting observations with leverage one:
##   2, 5, 8, 11, 15, 18, 21, 24, 25, 28, 31, 34
```


Residuals vs Fitted
Residuals
Fitted values
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
Scale−Location
√|Standardized residuals|
Fitted values
Residuals vs Leverage
Standardized residuals
Cook's distance
Leverage